

Proposed architecture and budget

Service	Initial configuration	Monthly	Why included
ECS on Fargate	2 tasks × 0.5 vCPU / 1 GiB	\$36.04	Runs the participant web app and the director–actor agent endpoint (ElevenLabs custom LLM) without EC2 management.
RDS PostgreSQL	db.t4g.small; Single-AZ; 20 GB gp3	\$25.66	Managed, backed-up store for sessions, Prolific IDs, completion codes, and encounter metadata.
Application Load Balancer	1 public ALB across 2 AZs	\$16–22	Browser-trusted HTTPS for the app and the agent endpoint ElevenLabs calls; health checks; replaces ngrok from the pilot.
VPC egress and IPv4	1 NAT; 2 public IPv4; ~10 GB processed	\$40–44	Private workloads with controlled outbound access to ElevenLabs and the LLM API.
S3 (recordings & artifacts)	3 private versioned buckets; ~100–120 GB (video) growing	\$3–8	Durable storage for encounter video + audio, transcripts, steering logs (audit trail), and Terraform state — the study's rater-facing data. Webcam video uploads go browser → presigned S3 URL, never through the app servers or NAT.
CloudFront	1 distribution over the recordings bucket	≈\$0	Rater video streaming; the 1 TB/month free-tier egress covers ~3 raters × 400 videos, avoiding ~\$0.09/GB S3 egress.
Secrets Manager	Approximately 6 secrets	\$2.40	ElevenLabs, LLM, and DB credentials out of code and task definitions.
KMS	2 customer-managed keys	\$2.00	Auditable encryption for recordings and database — supports the IRB data-management plan.
Route 53	1 hosted zone	\$0.50	Stable DNS for the app and agent endpoint.
ECR	2 image repos	\$0.32	Immutable release-SHA images; the agent version used in each study wave is auditable.
CloudWatch and SNS	14-day logs, alarms, notification topic	\$0–5	Failure visibility during live participant sessions; alerts before Prolific complaints arrive.
ACM, EventBridge, SQS, IAM, Budgets	TLS, schedules, DLQs, access controls, alerts	≈\$0	Core security and operational controls at low volume.
EXPECTED TOTAL	Study-scale pilot + Phase 1 collection	\$128–148	Plan at \$160 to absorb usage variance.

REQUESTED MONTHLY BUDGET (AWS)	EXPECTED MONTHLY USE	EXPECTED DAILY USE
\$160 / month	\$128–148 / month	\$4.30–4.90 / day

AWS covers hosting only. The dominant study costs are external usage: ElevenLabs voice minutes and LLM tokens (next page), plus Prolific participant payments, which are study costs rather than infrastructure.

Current phase note (Aug 2026): during the agent pilot, the platform runs on AWS App Runner instead of the full stack above — approximately \$45–55/month while running, ~\$2/month paused between sessions, plus ECR/Secrets/CloudWatch at a few dollars. The table above is the production configuration for Phase 1 data collection; the \$160 budget request covers either configuration.

Why each service is helpful

Application delivery

- **ECS/Fargate** runs the two containers — the participant-facing web app (consent, instructions, encounter launch, completion codes) and the director–actor agent server that ElevenLabs calls each turn — without server patching or capacity management.
- **The ALB and ACM** give the agent endpoint a stable, browser-trusted HTTPS URL. The pilot's ngrok tunnel is not acceptable for live participant traffic: it adds latency, drops, and an uncontrolled third party.
- **ECR** ties each deployment to immutable release-SHA images, so the exact agent version used in every study wave is auditable — a methods-section requirement, not just ops hygiene.

Managed data services

- **RDS PostgreSQL** stores sessions, Prolific PID mapping, scenario/variation assignment, and encounter metadata, with automated backups separate from application containers.
- **S3** holds the study's primary data — encounter video, audio, transcripts, and the director's steering logs — versioned and encrypted; raters and the Phase-3 scorer read from here. Video dominates: ~800 encounters × 10–12 min of 720p webcam ≈ 100–120 GB. Capture happens in-browser (MediaRecorder) and uploads via presigned URLs directly to S3, so video never transits the app servers or the NAT gateway.
- **CloudFront** fronts the recordings bucket for rater review; its 1 TB/month free-tier egress absorbs the streaming that would otherwise bill ~\$0.09/GB from S3. Access stays restricted via signed URLs.
- **ElastiCache is deliberately omitted** at this scale (≤100 concurrent participants is far below what Postgres plus the app layer can handle); it can be added later without rework if concurrency grows.

Security and operations

- **Secrets Manager and KMS** keep ElevenLabs/LLM/DB credentials out of code and images and provide auditable encryption for participant recordings — supporting the IRB data-management plan (IRB0151104).
- **CloudWatch, SNS, EventBridge, and SQS** surface failures during live sessions (a dropped agent endpoint mid-encounter costs a participant payment) and retain failed jobs for diagnosis.
- **IAM and security groups** restrict who can read recordings and reach the database; the agent endpoint accepts traffic only from the ALB.

External usage dependencies (outside the AWS estimate)

Dependency	Recommendation
ElevenLabs Agents	The main platform cost: ≈\$0.08/min of conversation. Phase 1 = 400 encounters × ~10 min ≈ 4,000 min (≈\$320 at the metered rate); the Scale plan (\$299/mo, 3,738 min included) fits a collection month, Pro (\$99/mo, 1,238 min) fits pilot months. Budget ≈\$700–900 across Phases 1+4; confirm plan tier and concurrency (≥20 concurrent) before launch.
LLM API (director + actor)	Via the Cornell LiteLLM proxy with a study-scoped, budget-capped key. Director = Gemini Flash (decided Aug 2026); actor = Claude Fable 5 or Opus 5 (pilot A/B pending). ≈\$0.20–0.80 per encounter depending on actor choice → under \$700 across the study; outside the AWS estimate.
Prolific	Participant and rater payments are study costs, not infrastructure; budgeted in the IRB/grant, not here.
Qualtrics	Cornell institutional license; no marginal cost.
GitHub Actions	CI for image builds; free tier sufficient at this cadence.

Official pricing sources

Public pricing pages used for the August 3, 2026 estimate. Refresh during Terraform plan review — prices and organization-wide free-tier usage change.

- **Compute and data:** AWS Fargate; RDS for PostgreSQL
- **Networking and DNS:** Elastic Load Balancing; Amazon VPC; Route 53; ACM
- **Security and operations:** Secrets Manager; KMS; CloudWatch; AWS Budgets
- **Storage:** ECR; S3
- **Voice platform:** ElevenLabs Agents pricing (elevenlabs.io/pricing/agents)
- **LLM:** Anthropic API pricing via Cornell LiteLLM proxy

AWS line items mirror the AI-Ready Workforce staging estimate (July 30, 2026) where configurations are identical (RDS, ALB, NAT, Secrets Manager, KMS, Route 53, ECR); deltas: one fewer Fargate task class of workload, no ElastiCache, larger S3 for recordings.